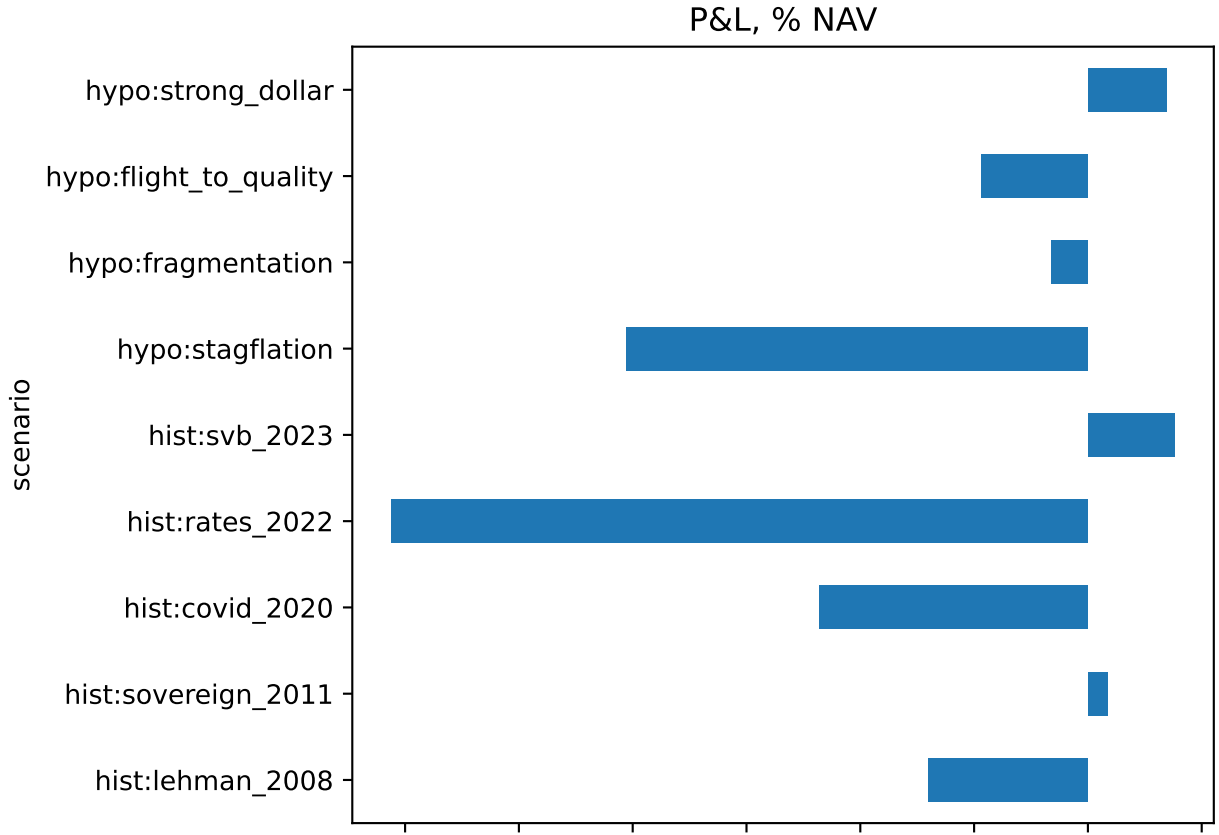
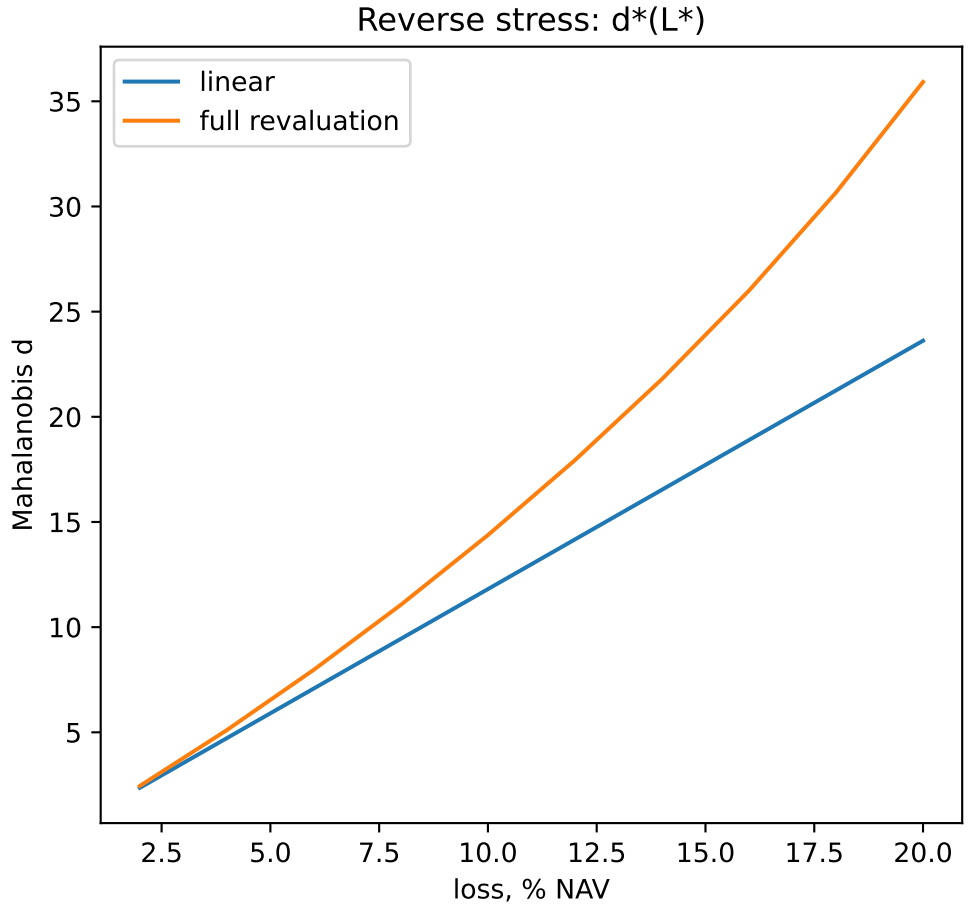
stress-engine: validation report (self-review)

Scope. Synthetic 100 MEUR book, nine lines, ten observable factors, instantaneous shocks (no theta, no expiry).
Status. Self-review by the author: not an independent validation.
Method. Full revaluation (two-pillar zero curve, Black-Scholes with ATM vol = VIX) against delta / delta-gamma Taylor, alert if the error exceeds 10 % of the P&L or 0.25 % of NAV. Partial shocks are completed by Gaussian conditional propagation (EWMA 0.97 + Ledoit-Wolf covariance, square-root-of-time scaling); Mahalanobis d is an ordinal scale.
Checks performed. Fiche control cases for bond and put (reproduced to 0.01-0.02), put-call parity, third-order slope of the Taylor remainder, QuantLib bond reference (gap 0.0098 % of nominal), asset-class cascade adds up, rolling out-of-sample backtest of the propagation (non-overlapping 10-day periods, PIT, KS, Berkowitz, Wilson).
Finding 1 (material). The Gaussian propagation is rejected by the KS test at 5 % for 94% of the factor/pilot pairs (18 tested, n = 424 periods each). The nominal 90 % band covers 90% to 98%: conservative, with fat tails. Use d as an ordinal scale only.
Finding 2. Reverse stress by bisection along the delta direction is not the exact optimum for a non-linear book (d at 10 % loss: 11.8 linear, 14.4 full revaluation).
Finding 3. Taylor order 2 raises the alert on 3 of 9 scenarios: do not use sensitivities for the crisis-sized moves of the book (options and convexity).
Finding 4. A Student-t propagation (nu = 5, fixed in advance, not tuned) lowers the KS rejection rate from 94% to 78%: a partial fix, tails are not the only problem.
Finding 5. The result depends on the covariance regime: the fragmentation scenario costs -0.6 % of NAV with the full sample, -0.0 % with stressed periods and -4.3 % since 2021.
Finding 6. The US corporate basket line is modelled with the BAA10Y spread; LQD regressed on it implies a duration of 3.6 (R2 0.10, 10-day tracking error 1.6 %): large basis risk.
Opinion. Fit for use as an exploratory stress tool with the conditions above; not fit to produce a probability.


P&L, % NAV
scenario
hypo:strong_dollar
hypo:flight_to_quality
hypo:fragmentation
hypo:stagflation
hist:svb_2023
hist:rates_2022
hist:covid_2020
hist:sovereign_2011
hist:lehman_2008
−12
−10
−8
−6
−4
−2
0
2


Reverse stress: d*(L*)
linear
full revaluation
Mahalanobis d
5
10
15
20
25
30
35
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
loss, % NAV